Discrete Diffusion PPO for Combinatorial Discrete Action Spaces

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Abstract

Formulation for discrete diffusion policy optimized with Proximal Policy Optimization (PPO) for large combinatorial discrete action spaces. The formulation is designed to avoid the continuous-action mismatch that arises when Gaussian diffusion is followed by thresholding or rounding. The diffusion variables and final actions are categorical throughout. PPO is applied to the stochastic denoising trajectory by treating the diffusion generation process as an augmented inner Markov process.

1 Problem Formulation

Let the environment be an MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}^N, P, r, \gamma), \tag{1}$$

where $\mathcal{S}$ is the state space, $\mathcal{A} = \{1, \ldots, K\}$ is a finite categorical action set, N is the number of simultaneous discrete decisions, P is the transition kernel, r is the environment reward, and $\gamma \in [0, 1)$ is the discount factor.

At environment time t,

$$s_t \in \mathcal{S}, \tag{2}$$

and the joint action is

$$\mathbf{a}_t = (a_t^1, \ldots, a_t^N), \qquad a_t^i \in \mathcal{A}. \tag{3}$$

Thus,

$$\mathbf{a}_t \in \mathcal{A}^N, \qquad |\mathcal{A}^N| = K^N. \tag{4}$$

For an offloading application,

$$K = 3, \qquad \mathcal{A} = \{\text{local}, \text{edge}, \text{cloud}\}. \tag{5}$$

With N simultaneous task decisions, the joint action space has size 3^N.

The policy objective is the standard discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{H-1} \gamma^t r_t\right], \tag{6}$$

where

$$s_{t+1} \sim P(\cdot \mid s_t, \mathbf{a}_t), \qquad r_t = r(s_t, \mathbf{a}_t, s_{t+1}). \tag{7}$$

2 Discrete Diffusion Process

The central design choice is that diffusion occurs directly in the categorical action space. There is no Gaussian action variable and no continuous-to-discrete thresholding operation.

For each action component,

$$x_k^i \in \mathcal{A}, \qquad k = 0, \dots, T. \tag{8}$$

The complete diffusion state is

$$\mathbf{x}_k = (x_k^1, \dots, x_k^N) \in \mathcal{A}^N. \tag{9}$$

2.1 Forward Process

Let

$$\pi = \left(\frac{1}{K}, \dots, \frac{1}{K} \right)^\top \tag{10}$$

be the uniform categorical distribution. For diffusion step k, define the transition matrix

$$Q_k = (1 - \beta_k) I + \beta_k \mathbf{1} \pi^\top, \qquad 0 < \beta_k < 1. \tag{11}$$

Equivalently,

$$Q_k(j, \ell) = \begin{cases} 1 - \beta_k + \dfrac{\beta_k}{K}, & j = \ell, \\ \dfrac{\beta_k}{K}, & j \neq \ell. \end{cases} \tag{12}$$

The forward categorical diffusion is

$$q(x_k^i \mid x_{k-1}^i) = \mathrm{Cat}\left(x_k^i ; Q_k[x_{k-1}^i, :] \right). \tag{13}$$

Assuming conditional independence across action components in the forward process,

$$q(\mathbf{x}_k \mid \mathbf{x}_{k-1}) = \prod_{i=1}^{N} q(x_k^i \mid x_{k-1}^i). \tag{14}$$

As the number of diffusion steps increases, the distribution approaches the uniform categorical distribution. We therefore initialize generation from

$$x_T^i \sim \mathrm{Cat}\left(\frac{1}{K}, \dots, \frac{1}{K} \right), \tag{15}$$

independently for $i = 1, \dots, N$.

3 Reverse Discrete Diffusion Policy

The policy is parameterized by a neural denoiser

$$f_\theta(s_t, \mathbf{x}_k, k). \tag{16}$$

For each action component i, the network outputs K logits:

$$\ell_{\theta,i,k} = f_\theta^i(s_t, \mathbf{x}_k, k) \in \mathbb{R}^K. \tag{17}$$

The reverse transition is categorical:

$$p_\theta(x_{k-1}^i = j \mid \mathbf{x}_k, s_t, k) = \frac{\exp(\ell_{\theta,i,k,j})}{\sum_{m=1}^{K} \exp(\ell_{\theta,i,k,m})}. \tag{18}$$

The joint reverse transition is factorized across action components:

$$p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s_t, k) = \prod_{i=1}^{N} p_\theta(x_{k-1}^i \mid \mathbf{x}_k, s_t, k). \tag{19}$$

The denoising trajectory is

$$\tau_t^d = (\mathbf{x}_T, \mathbf{x}_{T-1}, \ldots, \mathbf{x}_0). \tag{20}$$

The final environment action is defined directly by

$$\mathbf{a}_t = \mathbf{x}_0. \tag{21}$$

Consequently,

$$\mathbf{a}_t \in \mathcal{A}^N \tag{22}$$

4 Denoising Trajectory Probability

Let $p_T(\mathbf{x}_T)$ denote the fixed initial diffusion distribution. The probability of a complete denoising trajectory is

$$P_\theta(\tau_t^d \mid s_t) = p_T(\mathbf{x}_T) \prod_{k=1}^{T} p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s_t, k). \tag{23}$$

Therefore,

$$\log P_\theta(\tau_t^d \mid s_t) = \log p_T(\mathbf{x}_T) + \sum_{k=1}^{T} \log p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s_t, k) \tag{24}$$

$$= C + \sum_{k=1}^{T} \sum_{i=1}^{N} \log p_\theta(x_{k-1}^i \mid \mathbf{x}_k, s_t, k), \tag{25}$$

where $C = \log p_T(\mathbf{x}_T)$ is independent of θ.

This factorization makes the trajectory log-probability tractable without enumerating all K^N joint actions.

5 PPO Formulation

The marginal probability of the final action is

$$\pi_\theta(\mathbf{a} \mid s) = \sum_{\tau^d:\, \mathbf{x}_0 = \mathbf{a}} P_\theta(\tau^d \mid s), \tag{26}$$

which is generally intractable because many diffusion trajectories may lead to the same final action.

We therefore treat the diffusion generation process as an augmented inner stochastic process and apply PPO to the sampled denoising trajectory.

The trajectory likelihood ratio is

$$\rho_t(\theta) = \frac{P_\theta(\tau_t^d \mid s_t)}{P_{\theta_{\text{old}}}(\tau_t^d \mid s_t)}. \tag{27}$$

Since the initial distribution p_T is fixed, it cancels:

$$\rho_t(\theta) = \prod_{k=1}^{T} \frac{p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s_t, k)}{p_{\theta_{\text{old}}}(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s_t, k)}. \tag{28}$$

Equivalently,

$$\log \rho_t(\theta) = \sum_{k=1}^{T} \sum_{i=1}^{N} \Big[\log p_\theta(x_{k-1}^i \mid \mathbf{x}_k, s_t, k) \tag{29}$$

$$- \log p_{\theta_{\text{old}}}(x_{k-1}^i \mid \mathbf{x}_k, s_t, k) \Big]. \tag{30}$$

5.1 Advantage Estimation

Use a standard value function $V_\phi(s_t)$. Define the temporal-difference residual

$$\delta_t = r_t + \gamma(1 - d_t)V_\phi(s_{t+1}) - V_\phi(s_t), \tag{31}$$

where d_t indicates episode termination.

Generalized Advantage Estimation (GAE) is

$$\hat{A}_t = \sum_{\ell=0}^{H-t-1} (\gamma\lambda)^\ell \delta_{t+\ell}. \tag{32}$$

The return target is denoted by $\hat{R}_t$.

5.2 Clipped Actor Objective

The PPO objective is

$$L_{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta)\hat{A}_t, \text{clip}\left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon\right) \hat{A}_t \right) \right]. \tag{33}$$

The actor minimizes

$$\mathcal{L}_{\text{actor}} = -L_{\text{PPO}}. \tag{34}$$

5.3 Critic Objective

The critic is a conventional state-value estimator:

$$\mathcal{L}_{\text{critic}} = \frac{1}{2}\mathbb{E}_t \left[V_\phi(s_t) - \hat{R}_t \right]^2. \tag{35}$$

No diffusion model is required for the critic.

6 Policy-Gradient Interpretation

The environment reward depends on the final action $\mathbf{x}_0$. The expected return can be expressed as

$$J(\theta) = \mathbb{E}\left[R(s, \mathbf{x}_0)\right]. \tag{36}$$

Using the likelihood-ratio identity,

$$\nabla_\theta J = \mathbb{E}\left[R \nabla_\theta \log P_\theta(\tau^d \mid s) \right] \tag{37}$$

$$= \mathbb{E}\left[R \sum_{k=1}^{T} \nabla_\theta \log p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s, k) \right]. \tag{38}$$

Replacing the return with an appropriate advantage yields

$$\nabla_\theta J = \mathbb{E}\left[\hat{A}_t \sum_{k=1}^{T} \nabla_\theta \log p_\theta(\mathbf{x}_{k-1} \mid \mathbf{x}_k, s_t, k) \right]. \tag{39}$$

Thus the final environment reward can train every stochastic denoising transition through the augmented trajectory likelihood.

7 Network Architecture

A simple implementation uses a direct state encoder

$$h_t = f_\psi(s_t), \tag{40}$$

without an autoencoder.

The denoiser receives

$$(h_t, \mathbf{x}_k, k) \tag{41}$$

and produces $N \times K$ logits.

A Transformer is a natural implementation when interactions among the N discrete decisions are important:

$$(h_t, \mathbf{x}_k, k) \longrightarrow \text{Transformer/MLP} \longrightarrow N \times K \text{ logits}. \tag{42}$$

For the three-way offloading problem, the output at each denoising step contains $3N$ logits.

8 Rollout Buffer

The rollout buffer is on-policy. For each environment transition, store

$$\left(s_t, \mathbf{x}_T, \ldots, \mathbf{x}_0, r_t, s_{t+1}, V_\phi(s_t), \log P_{\theta_{\text{old}}}(\tau_t^d \mid s_t)\right). \tag{43}$$

Equivalently, it is sufficient to store the diffusion states needed to recompute the old and new transition log-probabilities, together with the environment transition and old trajectory log-probability.

9 Training Algorithm

1. Initialize actor parameters θ and critic parameters ϕ.
2. For each environment state s_t, sample
$$x_T^i \sim \text{Cat}(1/K, \ldots, 1/K).$$
3. For $k = T, T-1, \ldots, 1$, sample
$$\mathbf{x}_{k-1} \sim p_{\theta_{\text{old}}}(\cdot \mid \mathbf{x}_k, s_t, k).$$
4. Set
$$\mathbf{a}_t = \mathbf{x}_0$$
and execute the discrete joint action in the environment.
5. Store the complete denoising trajectory and environment transition.
6. Compute GAE advantages $\hat{A}_t$ and return targets $\hat{R}_t$.
7. During PPO epochs, recompute the new trajectory log-probability
$$\log P_\theta(\tau_t^d \mid s_t) = \sum_{k=1}^{T} \sum_{i=1}^{N} \log p_\theta(x_{k-1}^i \mid \mathbf{x}_k, s_t, k) + C.$$
8. Compute the trajectory ratio
$$\rho_t = \exp\left[\log P_\theta(\tau_t^d \mid s_t) - \log P_{\theta_{\text{old}}}(\tau_t^d \mid s_t)\right].$$
9. Optimize the clipped actor loss and critic loss.
10. Repeat with a fresh on-policy rollout.

10 Computational Motivation

11 Theoretical Qualification

A conventional joint categorical policy over all combinations would require a distribution over K^N possible actions. The diffusion policy instead produces N categorical distributions with K classes at each of T denoising steps. With factorized reverse transitions, the number of output logits per denoising step is NK although the denoiser can model cross-component dependencies through its shared network and attention mechanism.

Thus diffusion provides a structured parameterization for large combinatorial action spaces rather than explicitly enumerating all joint actions. The trajectory ratio

$$\rho_t(\theta) = \frac{P_\theta(\tau_t^d \mid s_t)}{P_{\theta_{\text{old}}}(\tau_t^d \mid s_t)} \tag{44}$$

is the exact importance ratio for the sampled augmented diffusion trajectory.

It is generally *not* equal to the marginal final-action ratio

$$\frac{\pi_\theta(\mathbf{a}_t \mid s_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t \mid s_t)}, \tag{45}$$

because the marginal action probability requires summing over all diffusion trajectories that terminate at $\mathbf{a}_t$.

Therefore, the proposed method should be described precisely as PPO on the augmented denoising trajectory rather than as exact PPO on an explicitly computed marginal joint-action probability.

12 Experimental Comparison

Vary the action dimension N and compare:

1. **Factorized categorical PPO:**

$$\pi_\theta(\mathbf{a} \mid s) = \prod_{i=1}^{N} \pi_\theta(a_i \mid s).$$

2. **Autoregressive PPO:**

$$\pi_\theta(\mathbf{a} \mid s) = \prod_{i=1}^{N} \pi_\theta(a_i \mid a_{<i}, s).$$

3. **Discrete Diffusion PPO:**

$$\mathbf{x}_T \to \cdots \to \mathbf{x}_0 = \mathbf{a}.$$

The principal evaluation should examine performance, stability, sample efficiency, and computational cost as K^N grows. For an offloading environment with $K = 3$, useful action dimensions include progressively larger numbers of simultaneous task decisions.

13 Summary of the Final Formulation

The complete method can be summarized as

$$s_t \to \text{categorical diffusion policy} \to \mathbf{x}_0 = \mathbf{a}_t \to (r_t, s_{t+1}) \tag{46}$$

with

$$\mathbf{a}_t \in \mathcal{A}^N, \qquad |\mathcal{A}^N| = K^N \tag{47}$$

and PPO optimized using the augmented denoising trajectory ratio

$$\rho_t = \frac{P_\theta(\mathbf{x}_{T:0} \mid s_t)}{P_{\theta_{\text{old}}}(\mathbf{x}_{T:0} \mid s_t)}. \tag{48}$$

The central research hypothesis is that iterative discrete denoising provides a scalable and expressive parameterization for large combinatorial discrete action spaces while preserving the natural compatibility of the environment with discrete actions.